No dimension-only stability constant for approximate m-star-convexity

Candidate counterexample for arXiv:2207.08058, Problem 1

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Status: candidate counterexample, likely valid. Problem 1 has a negative answer as literally stated: a stability constant depending only on the dimension cannot be uniform in the parameter $m \in (0, 1]$. Any corrected constant must be at least $1/(1 - m)$.

1 Source and problem

The source is Geanina Maria Lăchescu and Ionel Rovenţa, *The Hardy-Littlewood-Pólya inequality of majorization in the context of ω-m-star-convex functions*, arXiv:2207.08058 and *Aequationes Mathematicae* 97 (2023), 523–535 [1].

For a convex set $C \subseteq \mathbb{R}^N$, the source calls $\Phi : C \to \mathbb{R}$ δ-ω-m-star-convex when

$$\begin{aligned}\Phi((1-\lambda)x + \lambda m y) \le {} & (1-\lambda)\Phi(x) + m\lambda\Phi(y) \\ & - m\lambda(1-\lambda)\omega(\|x-y\|) + \delta\end{aligned} \tag{1}$$

for all $x, y \in C$ and $\lambda \in (0, 1)$. The exact class is obtained by deleting δ.

Problem 1 asks whether every such Φ decomposes as $\Phi = \Psi + \Pi$, where Ψ is exactly ω-m-star-convex and

$$\|\Pi\|_\infty \le k_N \delta, \tag{2}$$

with a positive constant k_N depending only on the dimension N.

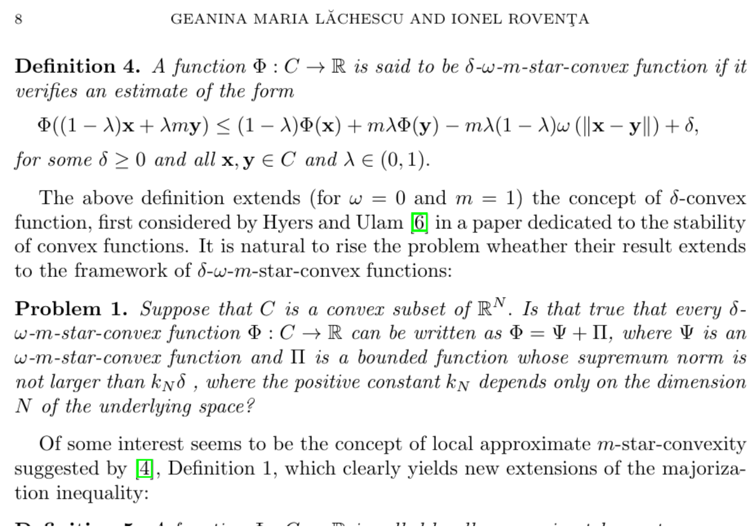
2 Proof intuition

For $m < 1$, the coefficients on the right of the star-convexity inequality sum to $1 - \lambda(1 - m) < 1$. At the origin this forces every exact function to be nonpositive. The allowed additive error δ, however, permits a positive constant of height $\delta/(1 - m)$. Therefore the distance from this approximate function to the exact class is at least $\delta/(1 - m)$. This diverges as m approaches one and rules out a constant depending only on dimension.

3 Quantitative obstruction and counterexample

Proposition 1. *Fix $N \ge 1$, $0 < m < 1$, and $\delta > 0$. Among all stability constants for Problem 1 at this fixed value of m, any constant valid for every convex domain and every zero-modulus approximately m-star-convex function must be at least*

$$\frac{1}{1-m}.$$



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Definition 4. *A function $\Phi : C \to \mathbb{R}$ is said to be δ-ω-m-star-convex function if it verifies an estimate of the form*

$$\Phi((1-\lambda)\mathbf{x} + \lambda m\mathbf{y}) \leq (1-\lambda)\Phi(\mathbf{x}) + m\lambda\Phi(\mathbf{y}) - m\lambda(1-\lambda)\omega(\|\mathbf{x}-\mathbf{y}\|) + \delta,$$

for some $\delta \geq 0$ and all $\mathbf{x}, \mathbf{y} \in C$ and $\lambda \in (0,1)$.

The above definition extends (for $\omega = 0$ and $m = 1$) the concept of δ-convex function, first considered by Hyers and Ulam [6] in a paper dedicated to the stability of convex functions. It is natural to rise the problem wheather their result extends to the framework of δ-ω-m-star-convex functions:

Problem 1. *Suppose that C is a convex subset of $\mathbb{R}^N$. Is that true that every δ-ω-m-star-convex function $\Phi : C \to \mathbb{R}$ can be written as $\Phi = \Psi + \Pi$, where Ψ is an ω-m-star-convex function and Π is a bounded function whose supremum norm is not larger than $k_N\delta$, where the positive constant k_N depends only on the dimension N of the underlying space?*

Of some interest seems to be the concept of local approximate m-star-convexity suggested by [4], Definition 1, which clearly yields new extensions of the majorization inequality:

Figure 1: Definition 4 and Problem 1 on page 8 of the source paper. The source explicitly says that k_N depends only on N.

Proof. Take $C = \mathbb{R}^N$, $\omega \equiv 0$, and define

$$\Phi(x) = c := \frac{\delta}{1-m} \qquad (x \in \mathbb{R}^N). \tag{3}$$

For arbitrary $x, y \in \mathbb{R}^N$ and $\lambda \in (0,1)$, the right-hand side of (1) is

$$\begin{aligned}(1-\lambda)c + m\lambda c + \delta &= c - \lambda(1-m)c + \delta \\ &= c + (1-\lambda)\delta \\ &\geq c.\end{aligned}$$

This equals or exceeds $\Phi((1-\lambda)x + \lambda my) = c$, so Φ is δ-0-m-star-convex.

Now let $\Psi : \mathbb{R}^N \to \mathbb{R}$ be exactly m-star-convex. Substituting $x = y = 0$ in its defining inequality gives

$$\Psi(0) \leq (1-\lambda+m\lambda)\Psi(0) = \big(1-\lambda(1-m)\big)\Psi(0).$$

Since $\lambda(1-m) > 0$, it follows that $\Psi(0) \leq 0$. Hence

$$\|\Phi - \Psi\|_\infty \geq |\Phi(0) - \Psi(0)| \geq \frac{\delta}{1-m}. \tag{4}$$

Thus no smaller universal factor can work at this m. □

Theorem 1. *There is no finite constant k_N depending only on N for which the conclusion of Problem 1 holds for all $m \in (0,1]$. Consequently, Problem 1 as stated has a negative answer.*

Proof. Fix N and suppose a finite dimension-only constant k_N exists. Choose $m \in (0,1)$ so close to one that

$$\frac{1}{1-m} > k_N.$$

The function Φ from Proposition 1 is an admissible approximate function, but (4) shows that every exact m-star-convex Ψ satisfies

$$\|\Phi - \Psi\|_\infty > k_N\delta.$$

Therefore no decomposition satisfying (2) exists. □

4 Verification, scope, and novelty

The example uses the full-dimensional domain $\mathbb{R}^N$ and the zero modulus. It therefore avoids both the domain-closure ambiguity in the source definition and any special choice of ω. The two essential checks are the exact calculation of the approximate defect and the origin substitution for exact functions; both appear explicitly above.

The result does not exclude a stability theorem with a constant $k_{N,m}$ allowed to depend on m. Rather, Proposition 1 shows that any such theorem covering the source's full class must satisfy

$$k_{N,m} \geq \frac{1}{1-m}.$$

If the notation k_N was intended to suppress hidden dependence on a fixed m, the result is a sharp necessary-parameter obstruction instead of a counterexample to that revised question.

Before promotion, the run indexes were searched using arXiv:2207.08058 and the core terms for approximate m-star-convexity and Hyers–Ulam stability. A bounded web/arXiv search used the exact Problem 1 sentence and close stability phrases. It found related results with extra hypotheses or parameter-dependent bounds, but no answer to the dimension-only ω-m problem. The source author's doctoral thesis, *Convex Analysis and Majorization Theory*, reproduces the problem in Section 2.3.4 [2]. This records the search bounds and is not an exhaustive novelty guarantee.

Human review is recommended. The main review issue is semantic rather than algebraic: confirm that "depends only on the dimension" is intended to mean uniformity in m.

References

[1] G. M. Lăchescu and I. Rovenţa, *The Hardy-Littlewood-Pólya inequality of majorization in the context of ω-m-star-convex functions*, Aequationes Math. 97 (2023), 523–535; arXiv:2207.08058. `https://arxiv.org/abs/2207.08058`.

[2] G. M. Lăchescu, *Convex Analysis and Majorization Theory*, doctoral thesis, University of Craiova, defended 2024 and confirmed 2025, Section 2.3.4. `https://ndsas.ucv.ro/results/`.